



Cold Spring Harbor Laboratory

Bioinformatics Training Hours

Hands-on Bulk ATAC-seq Analysis with CodeSpringLab

September 16th, 2025

Presented by:
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Bioinformatics Shared Resource (BSR)



Jupyter

Access to JupyterHub on CSHL remote computer (Elzar)

Request IT to create an Elzar account

- Open JupyterHub website link and select 'bamdev4' if prompted by a drop-down selection box:
<https://bamdev4.cshl.edu/jupyter>
- Enter your lab username and password, and press enter/return
- You'll immediately see contents in your home Elzar folder
- Select Kernel 'Bioinformatics Core' and a new notebook will open and you can type in the cell/box

Download and open CodeSpringLab package onto your home Elzar folder

- Open a new notebook as instructed above and copy+paste this in a cell/box:
`! git clone https://github.com/RadUtama/CodeSpringLab.git`
- Press return/enter and wait until the download is successful
- Go back to the tab/window where you can see your home contents and find and click 'CodeSpringLab'
- Then find and click 'bulkRNAseq' and then find and click 'bulkRNAseq.ipynb'
- bulkRNAseq notebook will open in a new tab, then select 'Kernel' > 'Change Kernel' > 'Bioinformatics Core'
- Click on refresh symbol button on top, below 'Kernel' menu before start working. Cheers

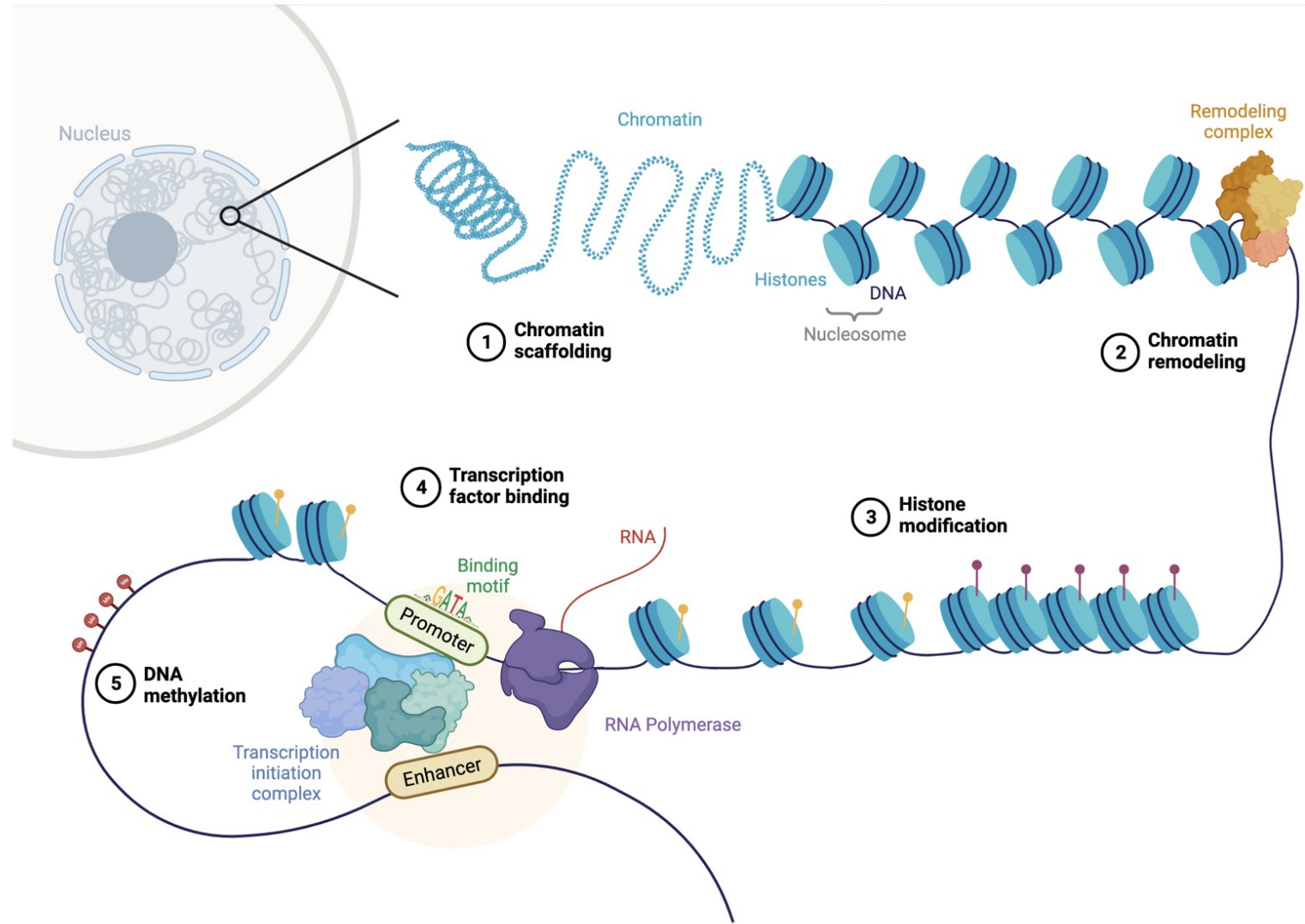
Galaxy to Jupyter

Galaxy to Jupyter

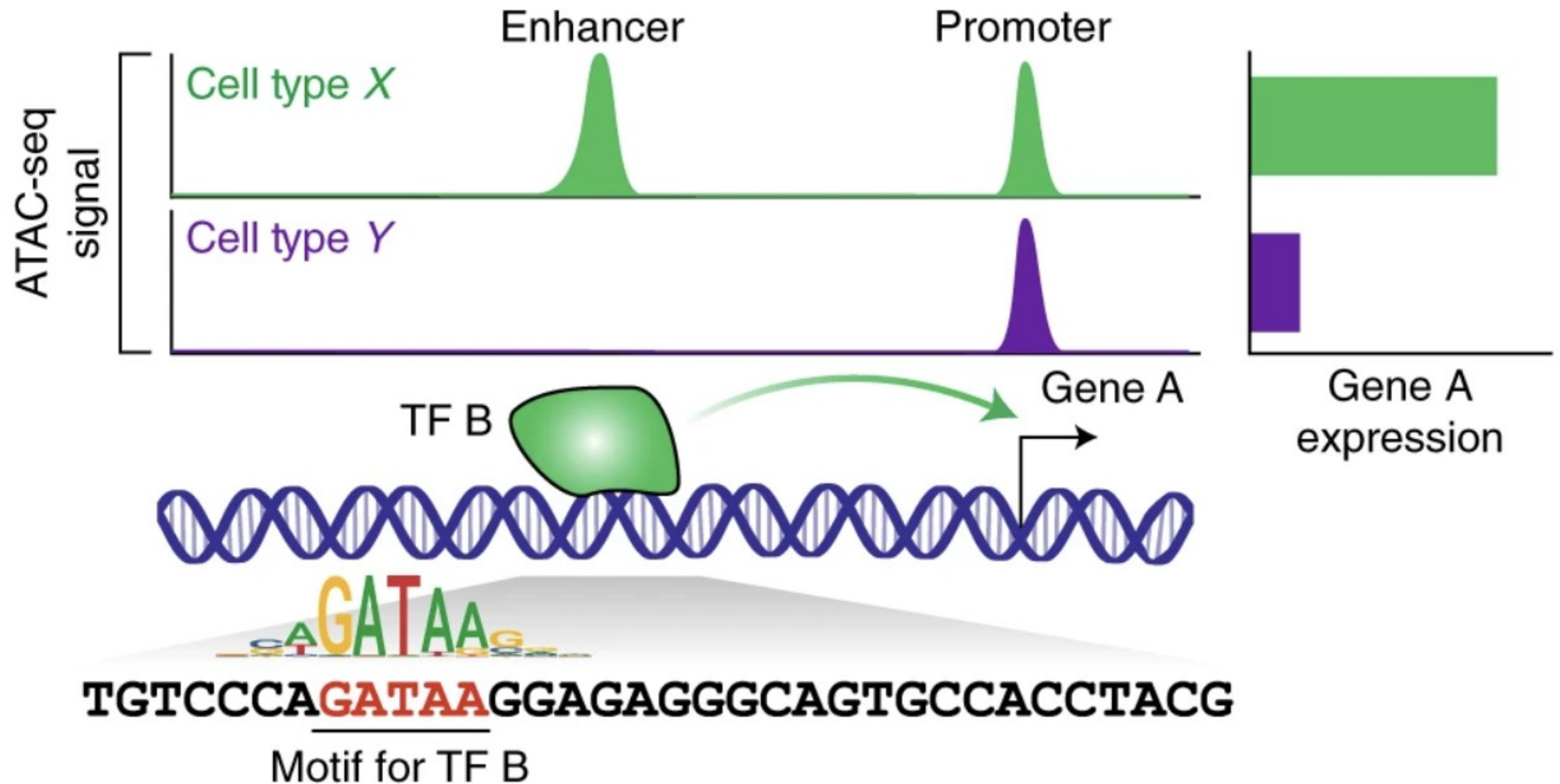
Why Jupyter ?

Features	Galaxy	Jupyter
Customizing analysis and parameters	limited	Yes
Exploring new tools	No	Yes
Regular tool updates	No	Yes
Server maintenance	Yes	No
Flexible data storage	No	Yes
User friendly	limited	Yes
Registration requirement	Yes	No
Coding requirement	No	Mostly No
More than 1 sample requirement	No	Yes
Tutorial requirement	No	Yes

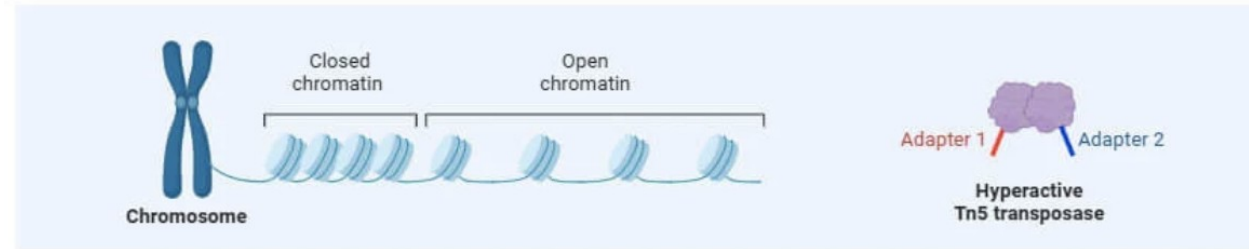
Intro to Bulk ATAC-seq



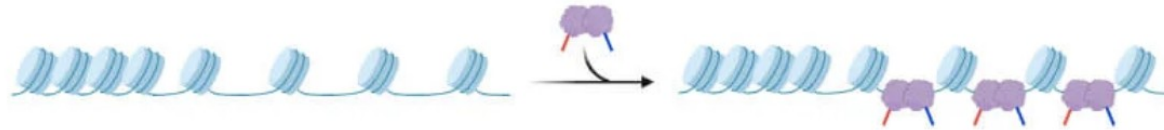
Intro to Bulk ATAC-seq



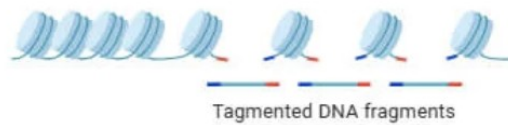
Intro to Bulk ATAC-seq



① Binding of Tn5 transposases to open chromatin



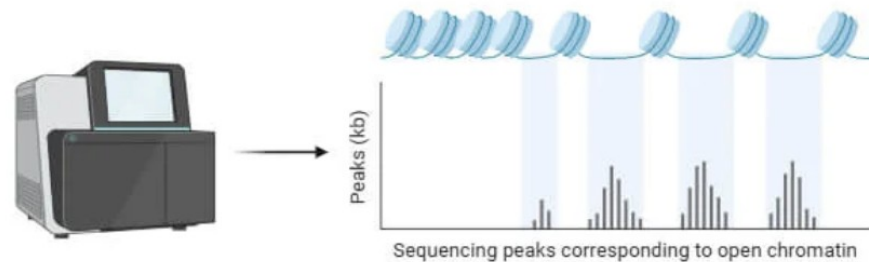
② Fragmentation and tagging (tagmentation) by Tn5 transposase



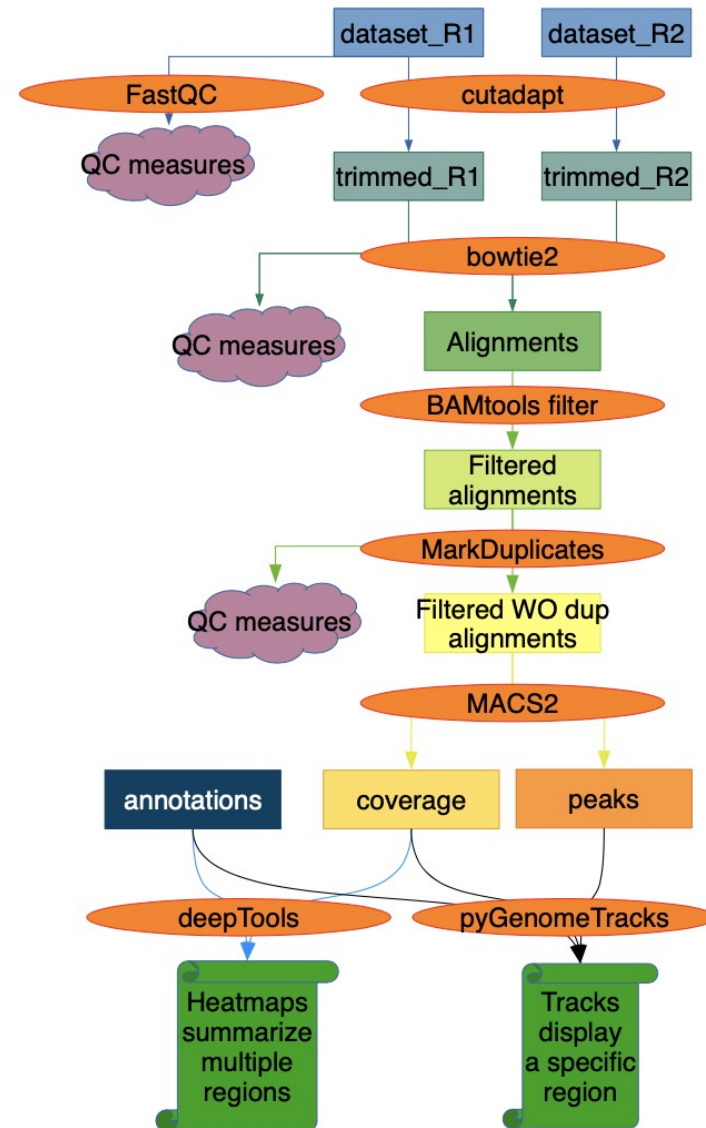
③ DNA fragment purification and PCR amplification



④ Next generation sequencing of PCR product and data analysis



Short Read Bulk ATAC-seq Analysis



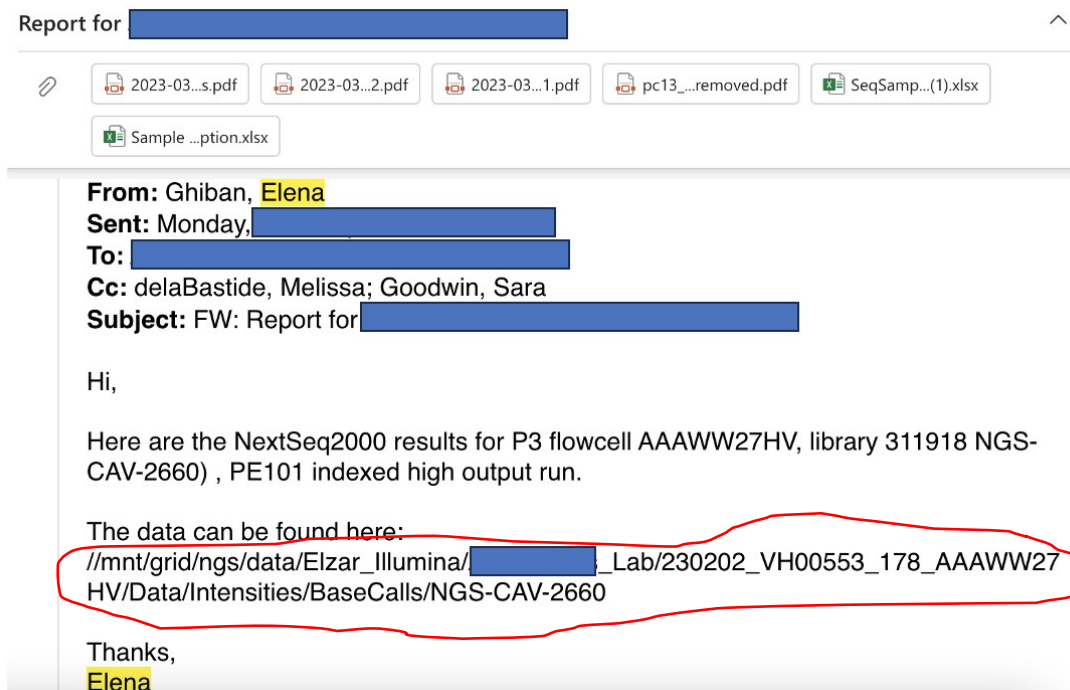
Short read

illumina

Long read



Input: [...]_R1/R2_001.fastq.gz sequencing file



Path/Location
of fastq files

Fastq file with reads:

```
@FORJUSP02AJWD1
CCGTCAATTCAATTAAGTTTAACTTGC GGCCGTACTCCCAGGCGGT
+
AAAAAAAAAAAAA::99@:::??@:::FFAAAAACCAA:::BB@?A?
```

Label

Sequence

Q scores (as ASCII chars)

Base=T, Q='!' = 25

QC: FastQC

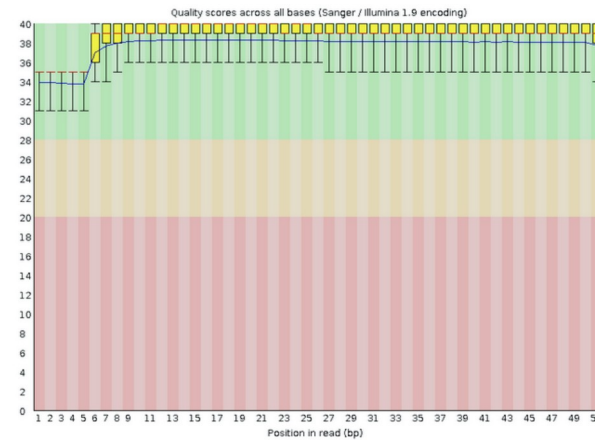
QC Interpretation:

https://hbctraining.github.io/Intro-to-rnaseq-hpc-salmon/lessons/qc_fastqc_assessment.html

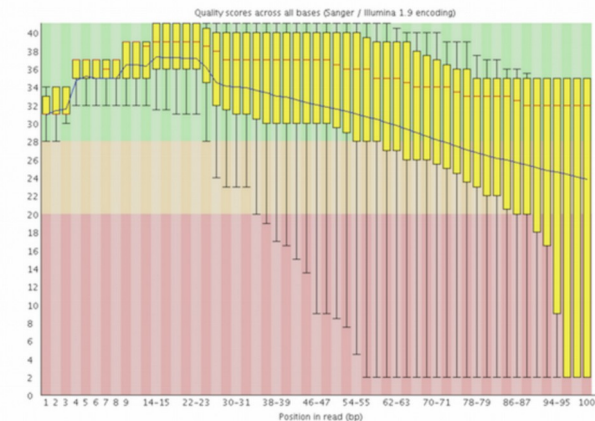
Summary

- ✓ Basic Statistics
- ✗ Per base sequence quality
- ! Per tile sequence quality
- ✓ Per sequence quality scores
- ✗ Per base sequence content
- ✗ Per sequence GC content
- ✓ Per base N content
- ✓ Sequence Length Distribution
- ✗ Sequence Duplication Levels
- ! Overrepresented sequences
- ✓ Adapter Content
- ! Kmer Content

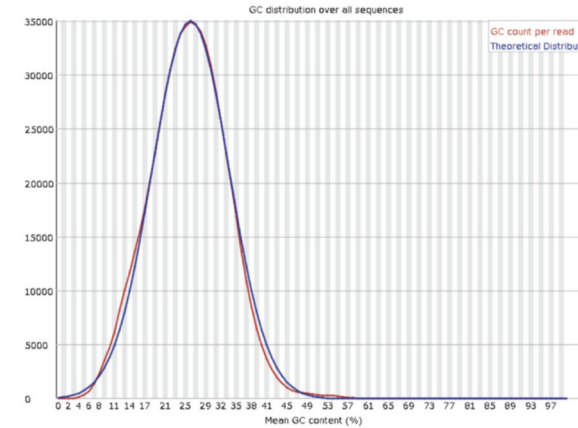
✓ Per base sequence quality



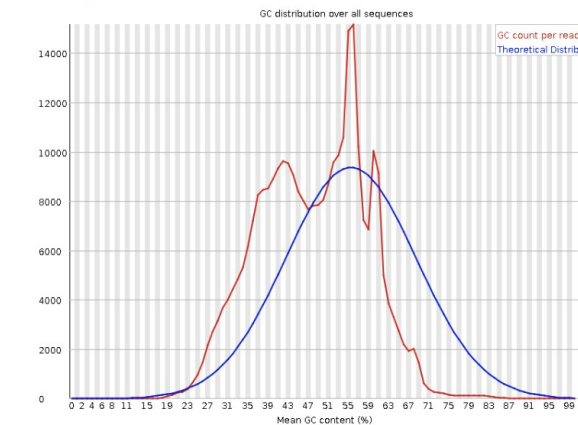
✗ Per base sequence quality



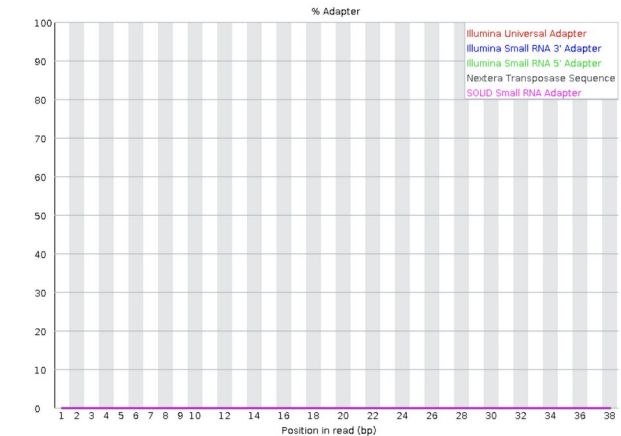
✓ Per sequence GC content



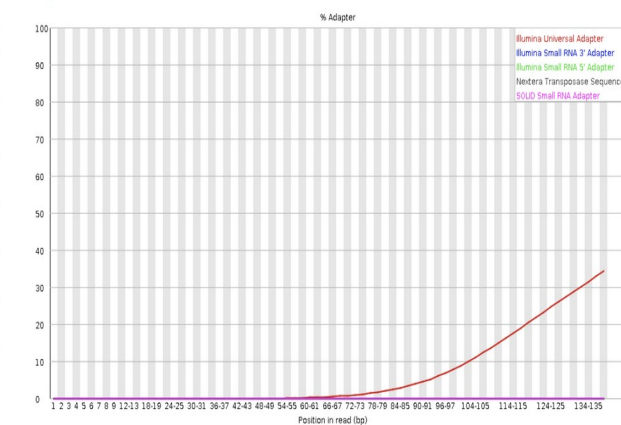
✗ Per sequence GC content



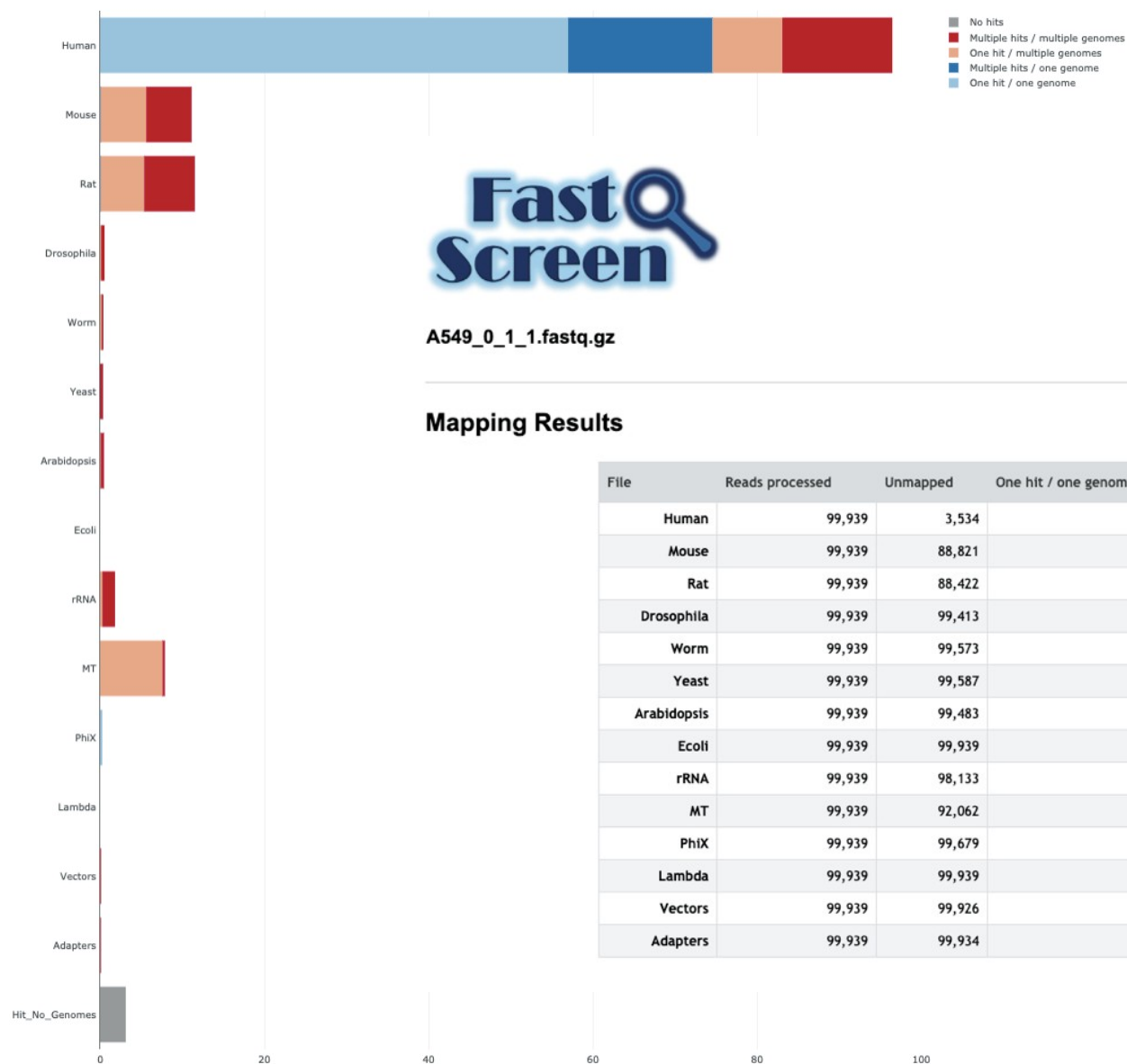
✓ Adapter Content



✗ Adapter Content



QC: Fastq-screen



File	Reads processed	Unmapped	One hit / one genome	Multiple hits / one genome	One hit / multiple genomes	Multiple hits / multiple genomes
Human	99,939	3,534	56,947	17,596	8,485	13,377
Mouse	99,939	88,821	42	6	5,558	5,512
Rat	99,939	88,422	34	15	5,302	6,166
Drosophila	99,939	99,413	0	0	129	397
Worm	99,939	99,573	1	0	237	128
Yeast	99,939	99,587	0	0	1	351
Arabidopsis	99,939	99,483	0	0	69	387
Ecoli	99,939	99,939	0	0	0	0
rRNA	99,939	98,133	0	0	275	1,531
MT	99,939	92,062	0	0	7,583	294
PhiX	99,939	99,679	260	0	0	0
Lambda	99,939	99,939	0	0	0	0
Vectors	99,939	99,926	0	0	5	8
Adapters	99,939	99,934	0	0	0	5

Alignment: Bowtie2

Bowtie2 end-to-end Alignment

Global/end-to-end alignment (default)

– Align the whole read

Option: --end-to-end

Can control how many gaps, how long, etc.

Better and faster for ATAC-seq short read (50-75 bp) with local alignment mode

Read: ATGCAGCTAGCTAGCTAGCTAGCT
|||||
Genome: ATGCAGCTAGCTAGC---CTAGCT 3nt insertion

Read: ATGCA--TAGCTAGCTAGCAAGCT
|||||
Genome: ATGCAGCTAGCTAGCTAGCTAGCT 2nt deletion
1 mismatch



Alignment: BAM output file

BAM file with aligned reads:

```
HWI-ST1145:74:C101DACXX:7:1102:4284:73714      16      chr20      190930      3      100M      *      0      0
      CCGTGTTTAAAGGTGGATGCGGTACCTTCCCAGCTAGGCTTAGGGATTCTTAGTTGGCCTAGGAAATCCAGCTAGTCCTGTCTCTCAGTCCCCCTCT
C      BBDCCDDCCDDDDCDDDDDDCDDCCDBC?DDDDDDDDDDDDDDCCDCDDDDDDDDDDCCCEDDDC?DDDDDDDDDDDDDDDDDDDDBDHFFFFDC@@
AS:i:-15      XM:i:3      XO:i:0      XG:i:0      MD:Z:55C20C13A9      NM:i:3      NH:i:2      CC:Z:=      CP:i:55352714      HI:i:0
```

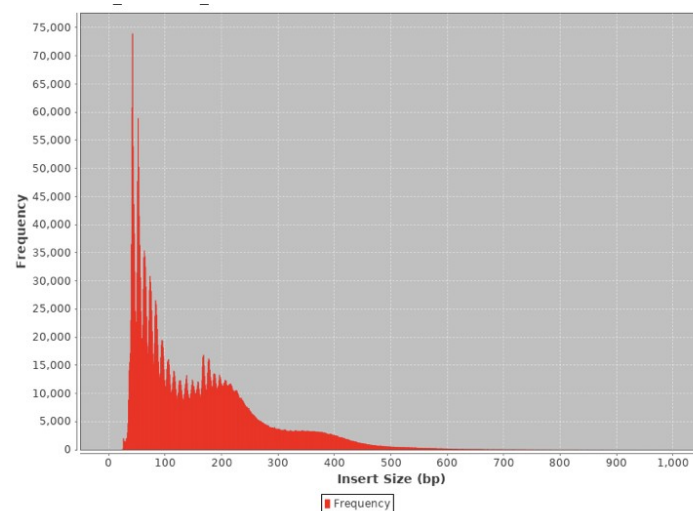
Mapping QC Rule of Thumb -> % Uniquely mapped reads

- 90 % = Ideal
- 80 % = Good
- 70 % = Acceptable
- 60 % = Borderline
- 50 % = Possible removal/resequence

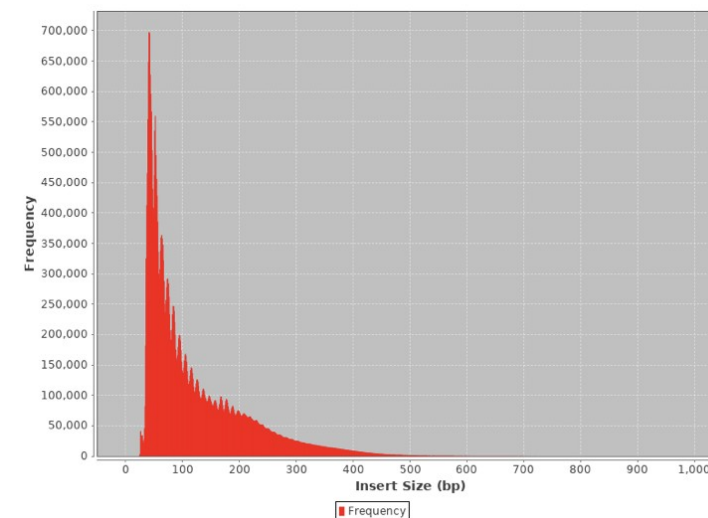
Alignment: Insert size QC

Insert size frequency

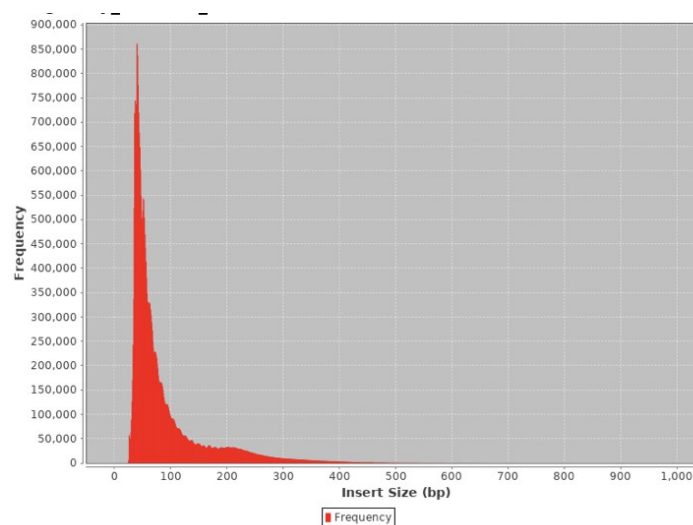
Good



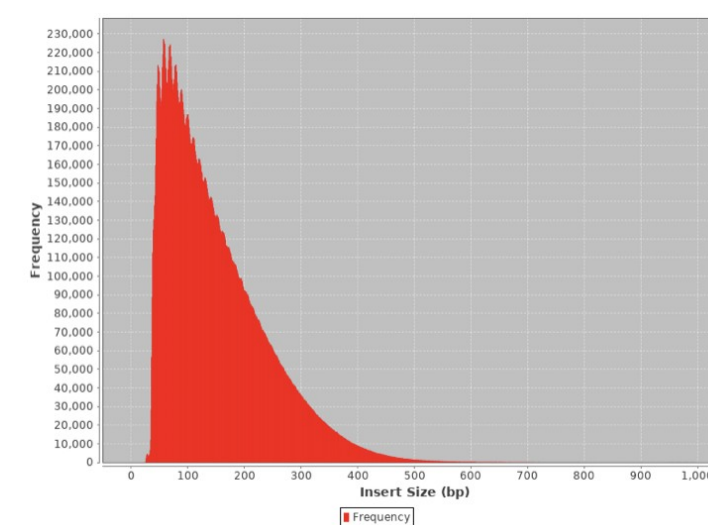
Noisy



Noisy

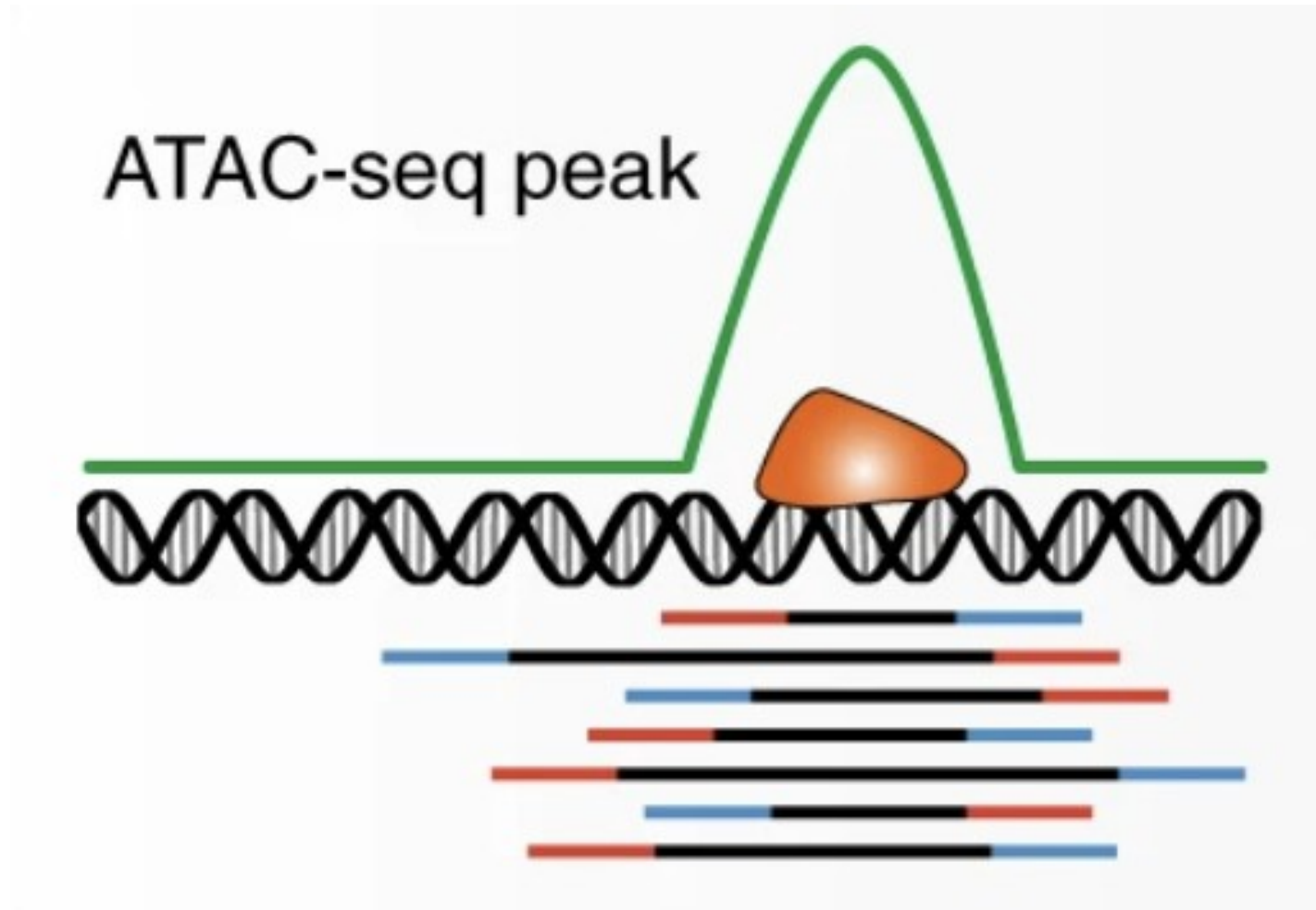


Failed



Quantification: DeepTools

DeepTools: bamCoverage (option `--normalizeUsing`)



Read Counts Normalization

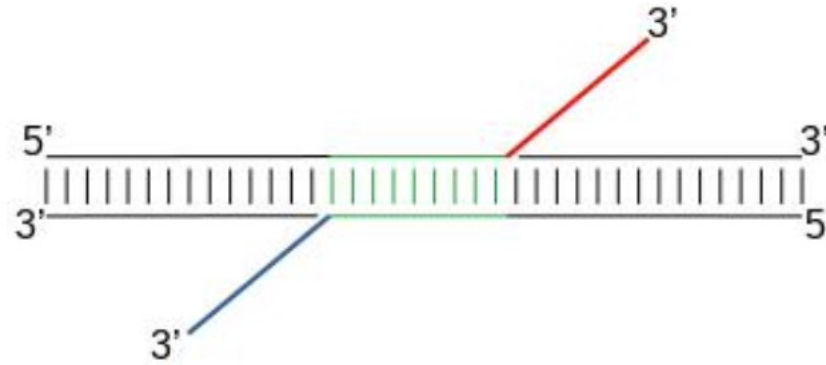
Raw count is **RAW** for a reason !

RPGC = count per bin / sequencing depth
Sequencing depth = total reads*read length/genome size

Spike-in normalization (E-Coli) for technical effect:
Set option **--scaleFactor**

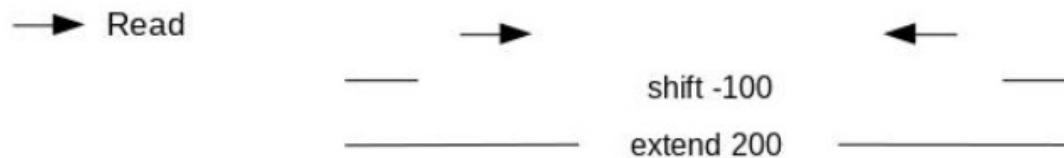
Normalization method	Description	Accounted factors	Recommendations for use
CPM (counts per million)	counts scaled by total number of reads	sequencing depth	gene count comparisons between replicates of the same samplegroup; NOT for within sample comparisons or DE analysis
TPM (transcripts per kilobase million)	counts per length of transcript (kb) per million reads mapped	sequencing depth and gene length	gene count comparisons within a sample or between samples of the same sample group; NOT for DE analysis
RPKM/FPKM (reads/fragments per kilobase of exon per million reads/fragments mapped)	similar to TPM	sequencing depth and gene length	gene count comparisons between genes within a sample; NOT for between sample comparisons or DE analysis
DESeq2's median of ratios [1]	counts divided by sample-specific size factors determined by median ratio of gene counts relative to geometric mean per gene	sequencing depth and RNA composition	gene count comparisons between samples and for DE analysis ; NOT for within sample comparisons
EdgeR's trimmed mean of M values (TMM) [2]	uses a weighted trimmed mean of the log expression ratios between samples	sequencing depth, RNA composition	gene count comparisons between samples and for DE analysis ; NOT for within sample comparisons

Peak Calling Analysis: MACS2



[Open image in new tab](#)

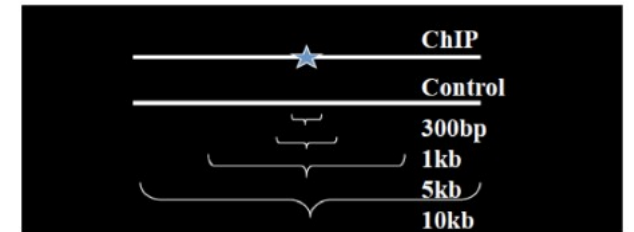
Figure 16: Nextera Library Construction



[Open image in new tab](#)

Figure 17: MACS2 options to get 100bp each side

Dynamic peak vs Background



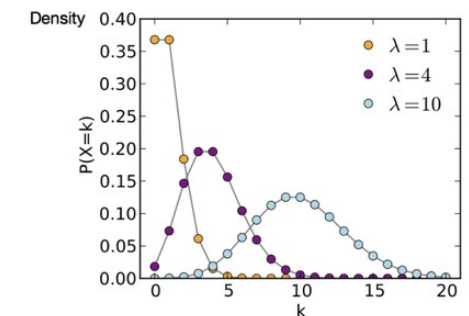
$$\lambda_{\text{local}} = \max(\lambda_{\text{BG}}, \lambda_{1\text{k}}, \lambda_{5\text{k}}, \lambda_{10\text{k}}).$$

$$P_{\lambda}(X=k) = \frac{\lambda^k}{k! * e^{-\lambda}}$$

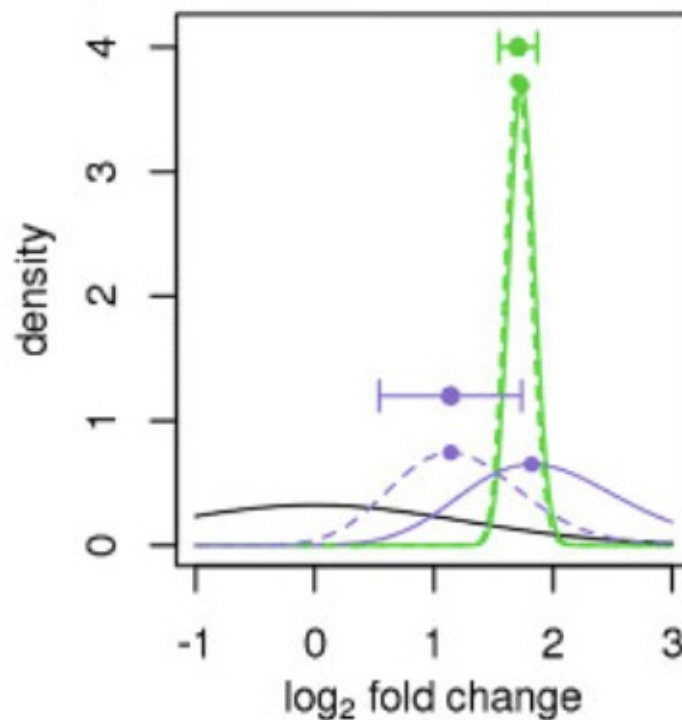
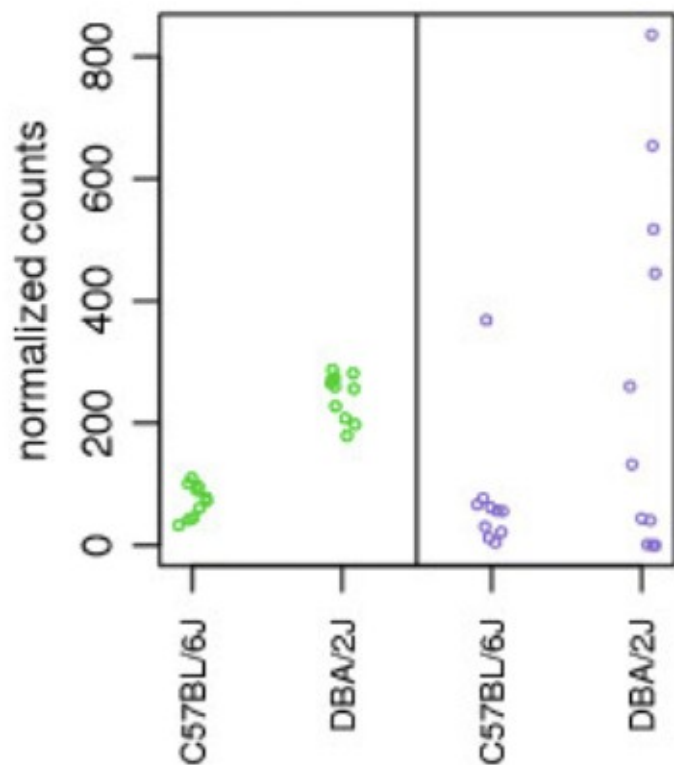
λ = mean = expected value = variance

$$\lambda = \frac{\text{total number of events (k)}}{\text{number of units (n) in the data}}$$

$$= \frac{\text{Read length (nt)} * \text{Total read number}}{\text{Effective genome length (nt)}}$$



Differential Peak Analysis: DiffBind + DESeq2

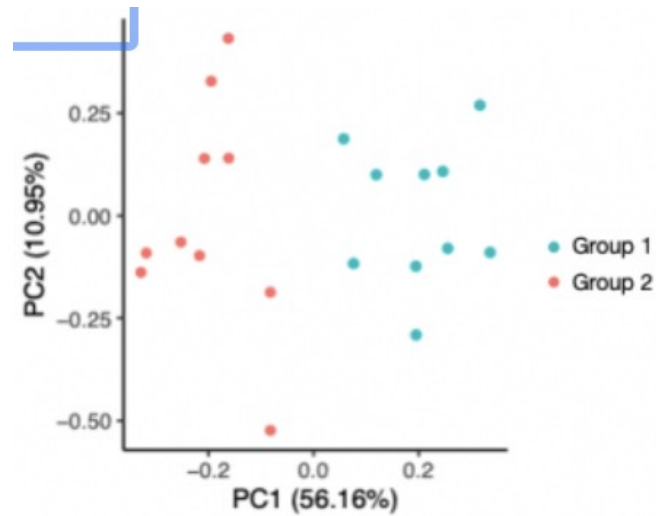


Output:

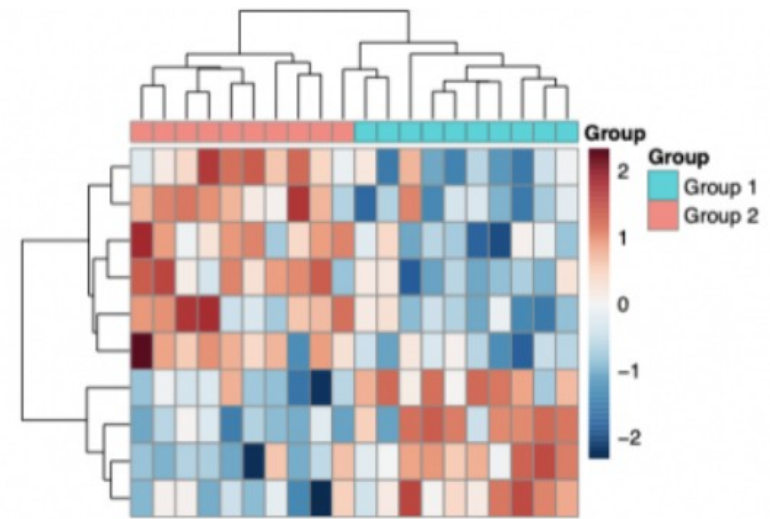
- baseMean
- logfoldchange (shrinkage)
 $\log_2(\text{normCounts.group2} / \text{normCounts.group1})$
- p-value (Wald test)
- FDR/adjusted p-value (Benjamini-Hochberg)

Visualization: PC, Heatmap, Volcano, Profilemap

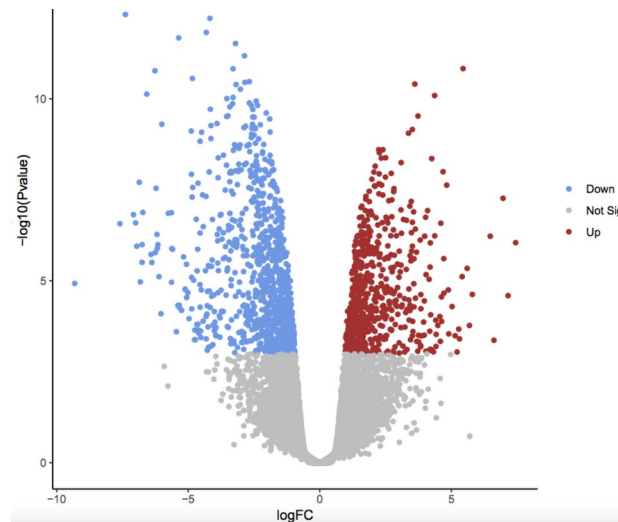
**Principal
Component
Dimensional
Reduction
Linear**



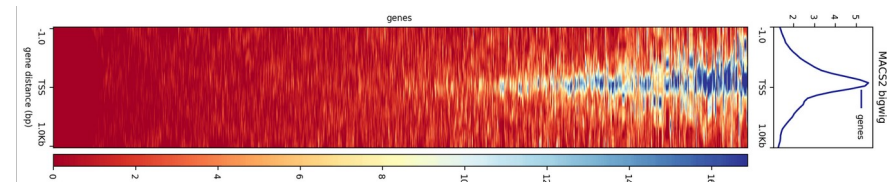
**Heatmap
Hierarchical
Clustering**



**Volcano
Logfoldchange
vs
-log(p-value)**



**Profilemap
Genes by TSS
vs
Binding signal**



Visualization: Genome tracks

UCSC Genome Browser, IGV, pyGenomeTracks

Continuous/density values

(normalized counts):

WIG: human readable

bigWig: compressed

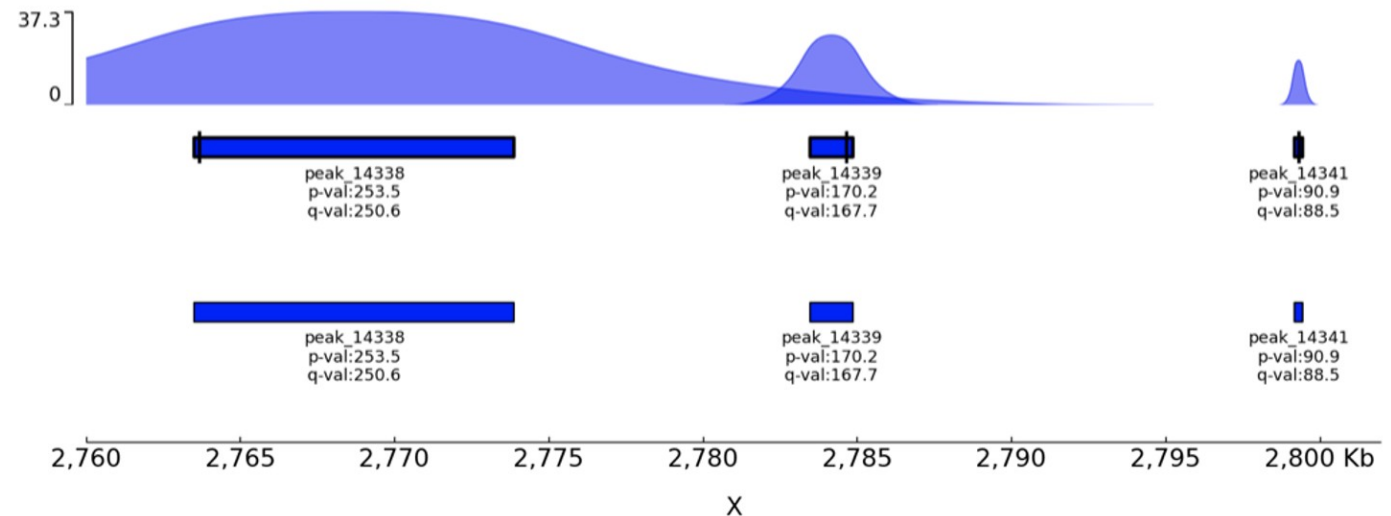
Ranged/discrete values

(peaks and summits):

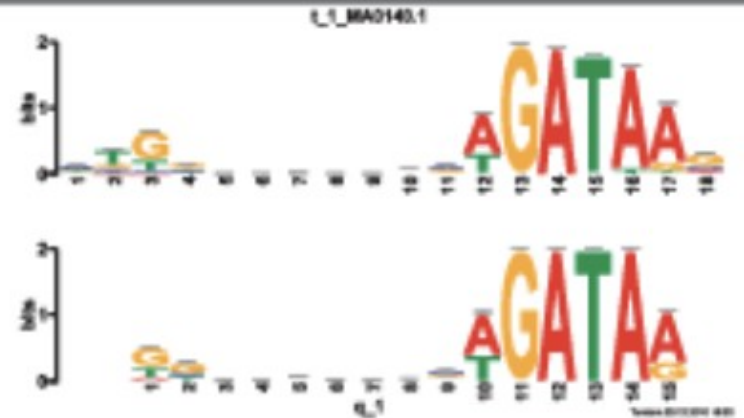
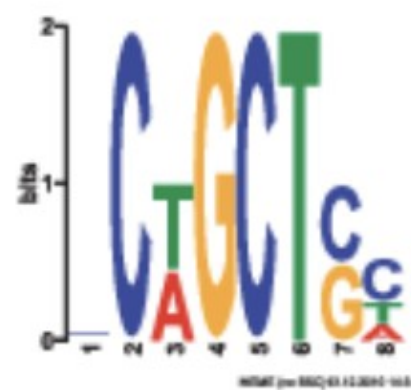
BED: Just boxes

bedGraph: Boxes with gradient color

narrowPeak: Boxes+gradient+summit loc



Motif analysis: MEME (not included)

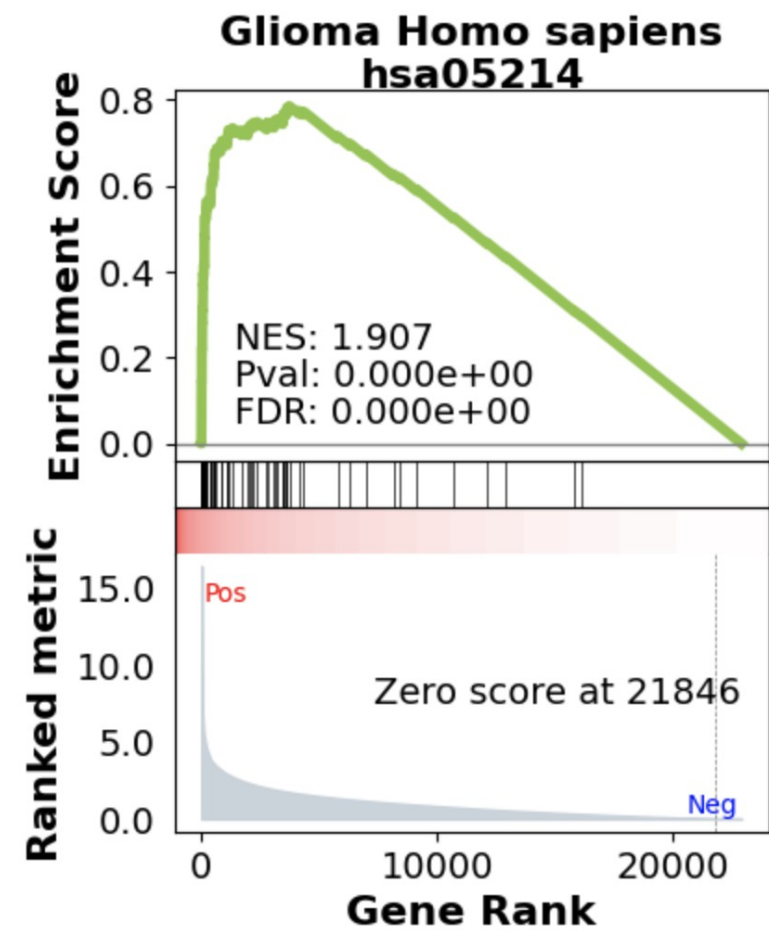
<i>algorithm</i>	<i>first motif</i>	<i>second motif</i>
MEME	 MEME	 MEME

Pathway Analysis: GSEA (not included)

Normalized Counts
(All samples)

Gene Set in a Pathway

Gene_set	Term	Overlap	P-value	Adjusted P-value	Old P-value	Old Adjusted P-value	Odds Ratio	Combined Score	
Hallmark_2020	IL-6/JAK/STAT3 Signaling	19/87	1.197225e-09	5.986123e-08	0	0	6.844694	140.612324	I
Hallmark_2020	TNF-alpha Signaling via NF-kB	27/200	3.220898e-08	5.368163e-07	0	0	3.841568	66.270963	BT
Hallmark_2020	Complement	27/200	3.220898e-08	5.368163e-07	0	0	3.841568	66.270963	FCN
Hallmark_2020	Inflammatory Response	24/200	1.635890e-06	2.044862e-05	0	0	3.343018	44.540108	L
Hallmark_2020	heme Metabolism	23/200	5.533816e-06	5.533816e-05	0	0	3.181358	38.509172	SLC22/



Thank you

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Free Training and Office hours:

Appointment by e-mail or this link:

<https://koalendar.com/e/bioinformatics-training-hours>
<https://calendly.com/utama/bioinformatics-office-hours>

One-on-One Hands-On Training:

Tuesdays, 10am - 1pm
Glaser Conference Room,
Matheson Building



Office Hours (in-person or virtual):

Tuesdays, 2pm - 5pm
Cushman Conference Room,
DeMatteis Building

